

Challenge (Habanero)

- Q1.** A hiker climbs 5 miles up a mountain and then descends 2 miles. What is the hiker's current position relative to the starting point, in miles, represented as an integer?
- (A) 3
(B) -3
(C) 5
(D) -5
(E) 2
- Q2.** The population of rabbits in a forest increases by 15 each month. If the current population is 25, what will be the population in 4 months, assuming no rabbits leave or die?
- (A) 25
(B) 65
(C) 45
(D) 85
(E) 105
- Q3.** A car travels 250 miles on 10 gallons of fuel. How many miles will it travel on 4 gallons of fuel, assuming the same fuel efficiency?
- (A) 25
(B) 50
(C) 100
(D) 125
(E) 150
- Q4.** The water level in a lake decreases by 2 inches every day due to evaporation. If the current water level is 10 inches above the minimum level, how many days will it take to reach the minimum level?
- (A) 2
(B) 3
(C) 4
- (D) 5
(E) 6
- Q5.** What is the next number in the pattern: 2, 5, 8, 11, 14, ...?
- (A) 15
(B) 17
(C) 18
(D) 20
(E) 22
- Q6.** A group of friends want to share some candy equally. If they have 48 pieces of candy and there are 8 friends, how many pieces of candy will each friend get?
- (A) 4
(B) 5
(C) 6
(D) 7
(E) 8
- Q7.** What is the value of x in the equation $x + 2 = 7$?
- (A) 3
(B) 5
(C) 7
(D) 9
(E) 11
- Q8.** A bicycle travels 12 miles in 2 hours. How many miles will it travel in 5 hours, assuming the same speed?
- (A) 20
(B) 25
(C) 30
(D) 35
(E) 40

Open Ended Questions (FRQ)

- Q9.** A waterfall drops 15 feet on the first day, 13 feet on the second day, and 11 feet on the third day. If this pattern continues, how many feet will the waterfall drop on the seventh day? Show your work and explain your reasoning.
- Q10.** A park ranger tracks the population of deer in a forest. The population starts at 50 deer and increases by 5 deer each month for 3 months, then decreases by 2 deer each month for 2 months. What is the final population of deer after these 5 months? Show your calculations and explain your answer.

Chef's Tips

- 1
- Tip 1: Encourage students to read the questions carefully and identify the key concepts being tested.
 - Tip 2: Remind students to use the correct units and formatting in their free-response answers.
 - Tip 3: Emphasize the importance of showing work and explaining reasoning in the free-response questions.